

Parallel Robots

EPFL – 2026

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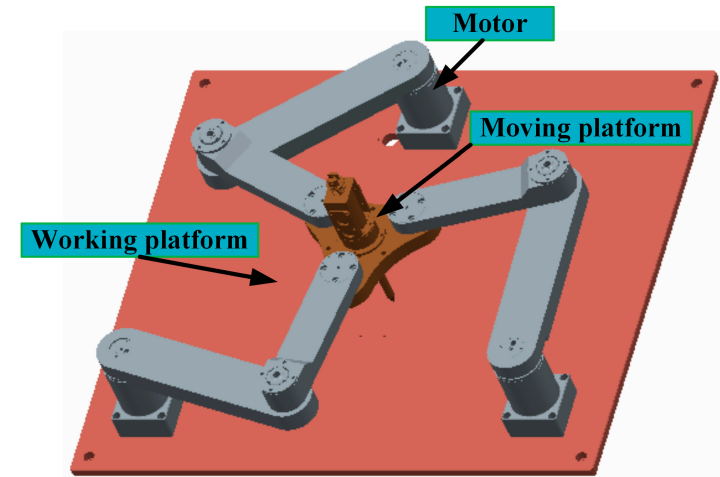


Introduction

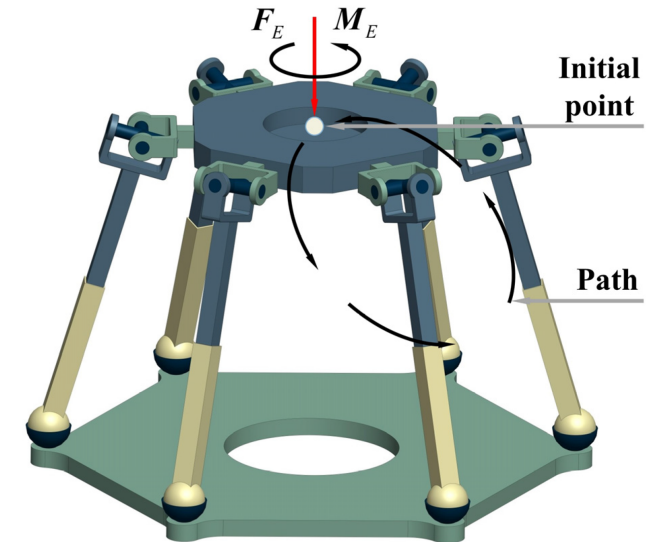
- According to J-P. Merlet:
 - “A generalized parallel manipulator is a closed-loop kinematic chain mechanism whose end-effector is linked to the base by several independent kinematic chains.”
 - Fully parallel if “the number of chains is strictly equal to the number of DOFs of the end-effector.”
 - → Not so important for us
- Characteristics of most robots we will be studying
 - At least 2 chains connecting the end-effector to the base
 - Only R, U, S and P joints (and parallelogram)
 - Often identical legs
 - Simple actuators distributed among legs
 - 1 sensor per actuator
 - When motors are locked (if possible), the robot cannot move

- *Active joint*: motorized
 - Underlined. For example. 3-RPR
- *Passive joint*: not motorized
 - Almost exclusive to parallel robots
- *Joint-space coordinate*: coordinate (or variable) associated to the motion of a given joint
- *Task-space coordinate*: coordinate associated to the DOFs of the end-effector (with some exceptions)
 - Also called Cartesian coordinate
- *Singularity*: configuration for which the mechanism loses the ability to move in (at least) one direction, or gains (at least) one uncontrollable motion
- *Assembly mode*: a solution to the forward geometric model (FGM)
 - Only for parallel robots
- *Working mode*: a solution to the inverse geometric model (IGM)

- Parallel robot topologies = the way joints are arranged
- How many joints per leg?
 - No clear answer in general. Why?
 - For 3-DOF planar fully parallel robots: 3 joints per leg
 - For 6-DOF spatial fully parallel robots: 6 joints per leg



<https://doi.org/10.3390/a11010007>

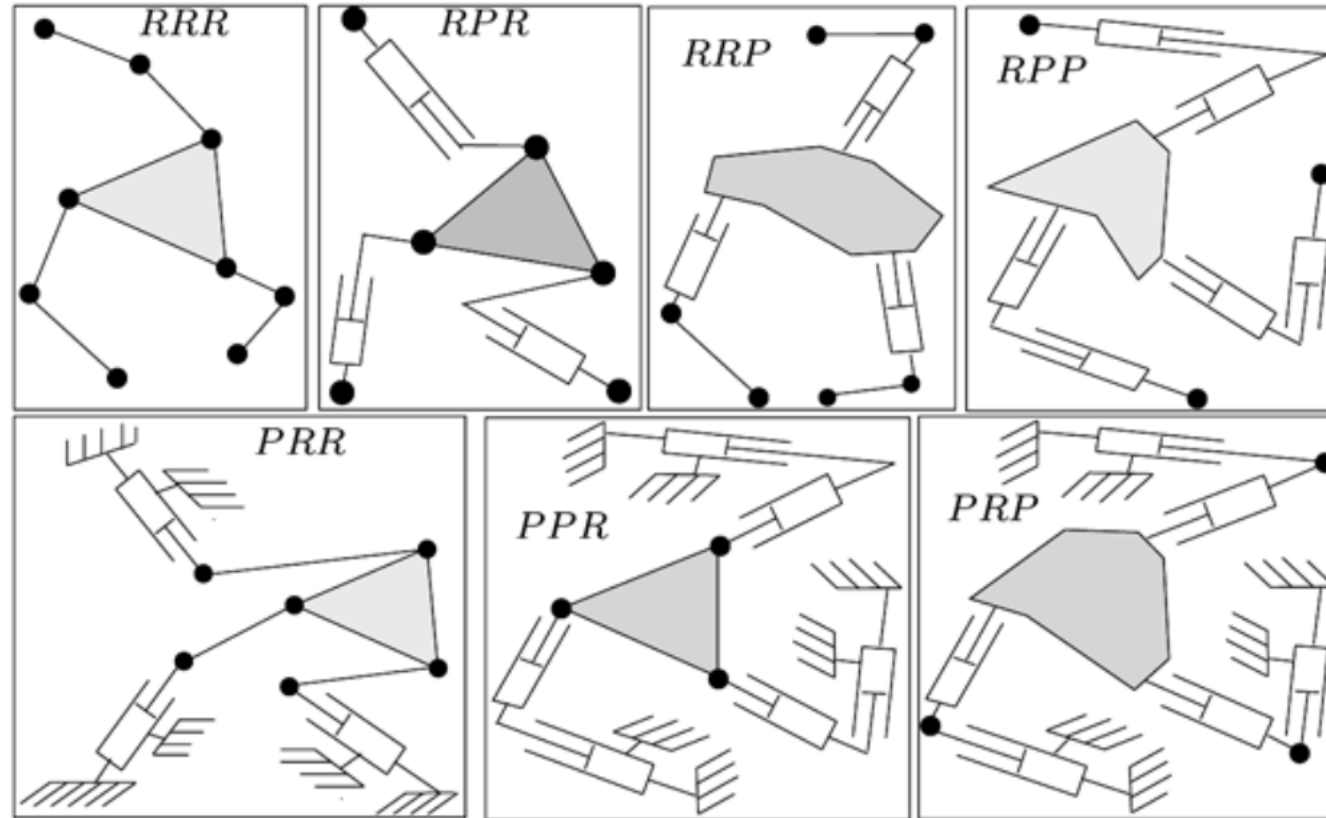


<https://doi.org/10.3390/app131810150>

- How to position a point in 2D?

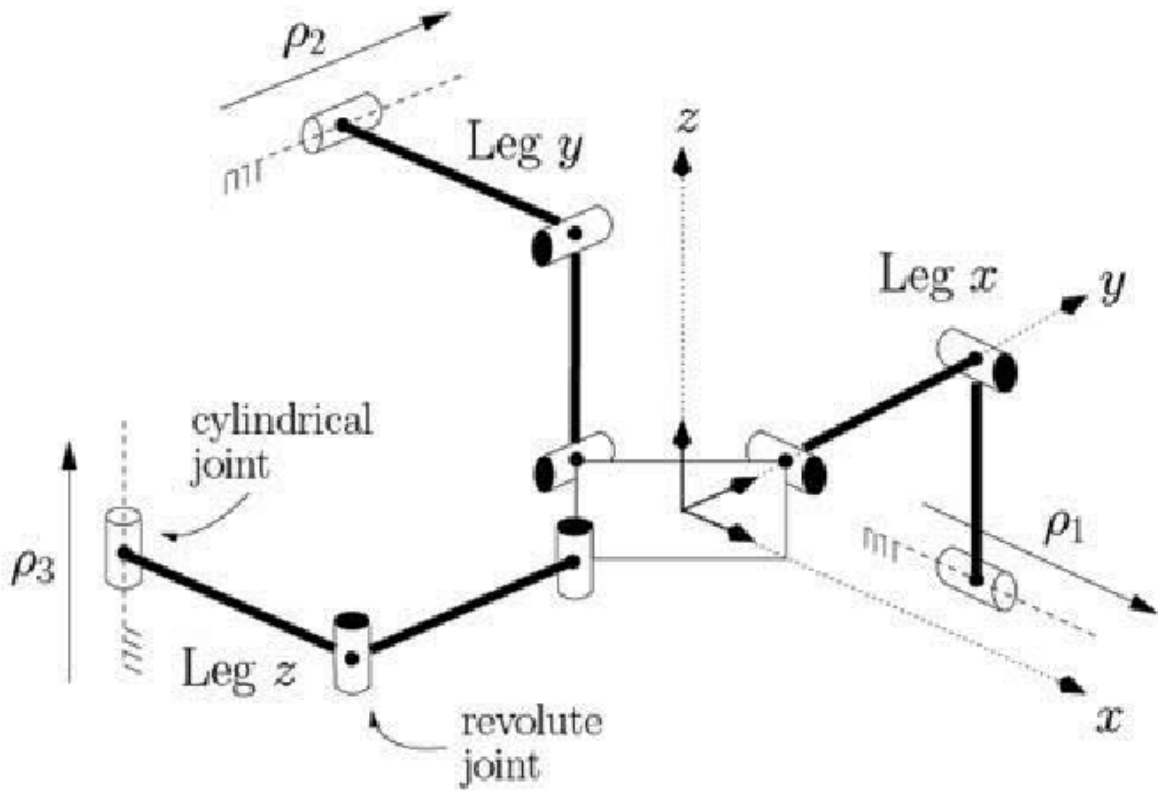
- How to position a point in 2D with constant orientation?

- How to do planar 3-DOF motion? (all solution)



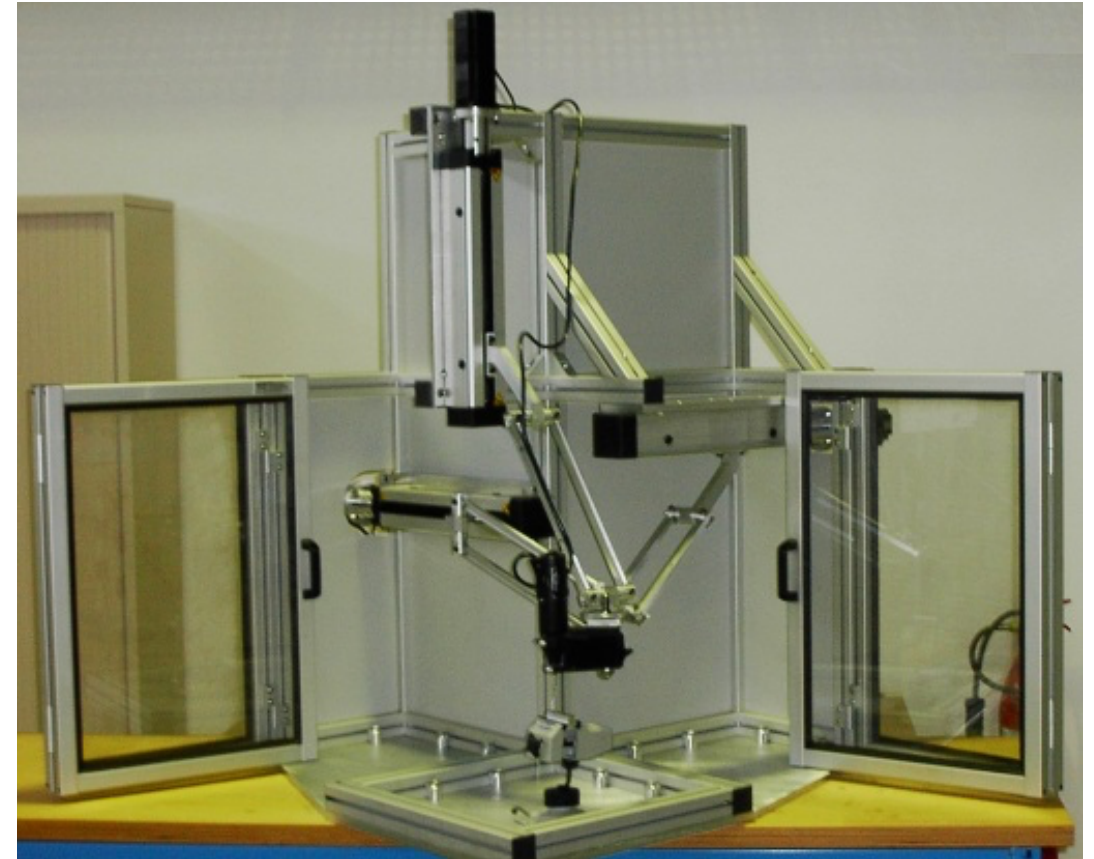
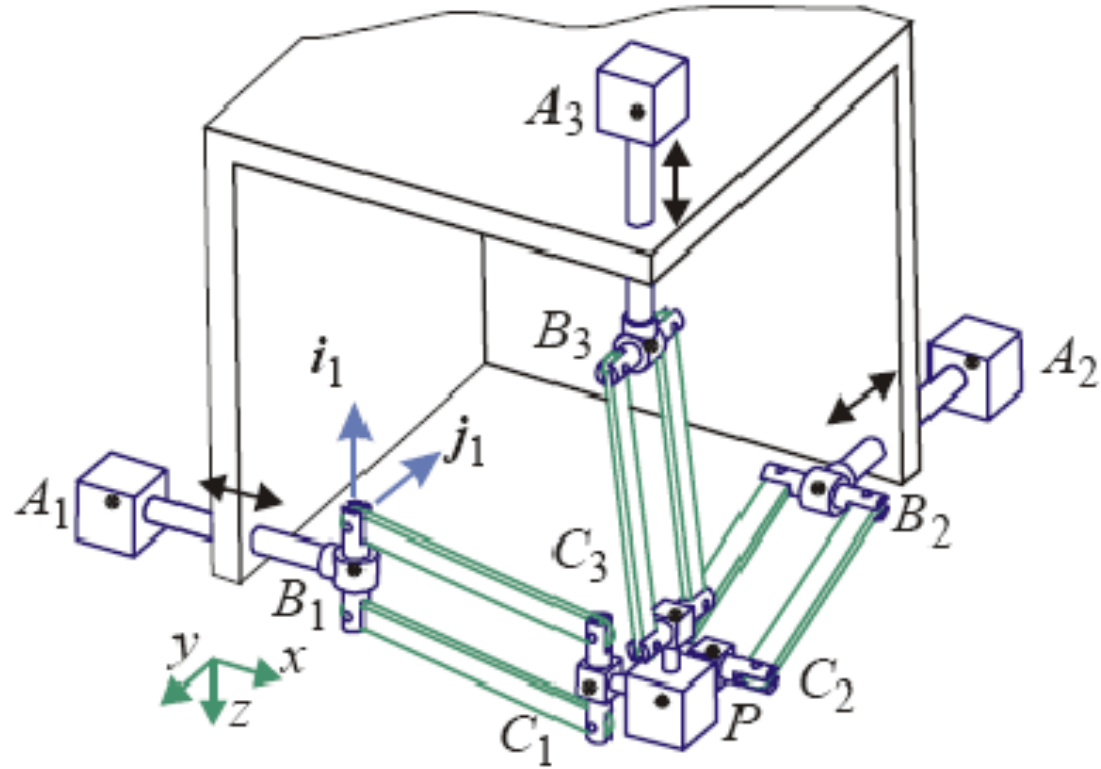
<https://doi.org/10.1007/1-4020-4133-0>

- How to do 3D translations?



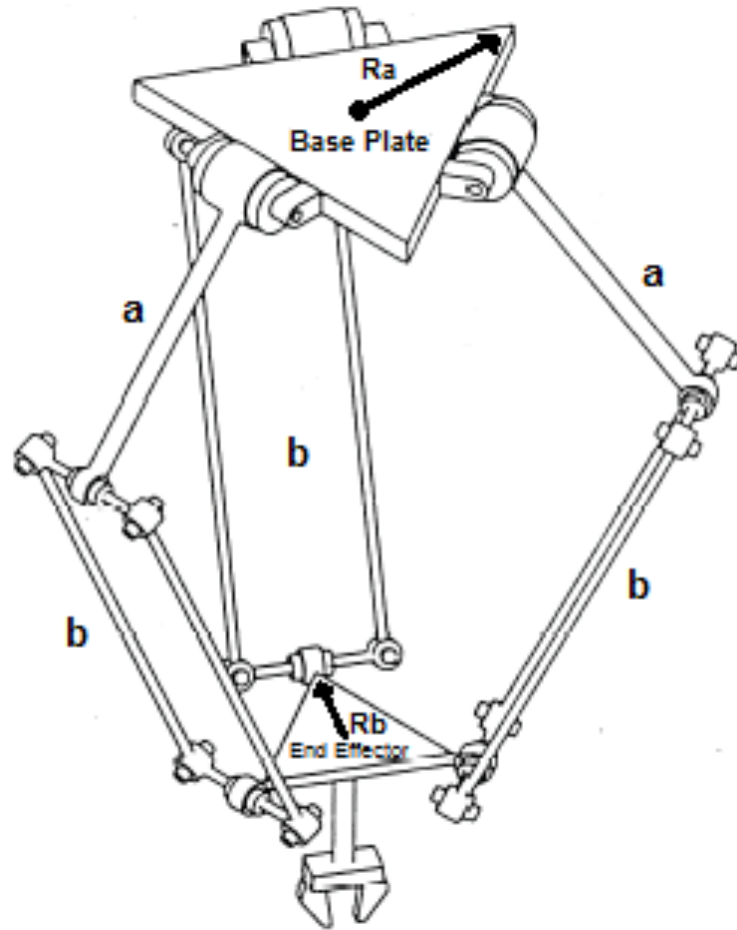
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- How to do 3D translations?



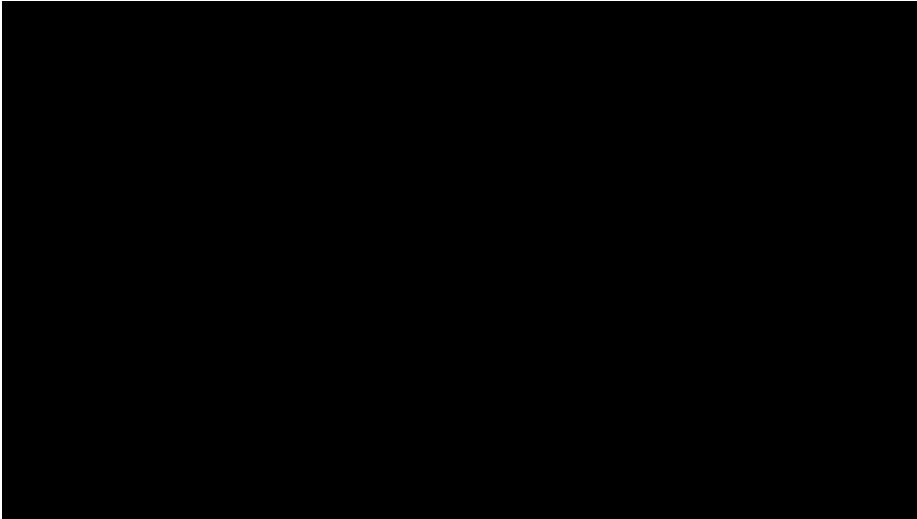
<https://www.parallemic.org/Reviews/Review011.html>

- How to do 3D translations?

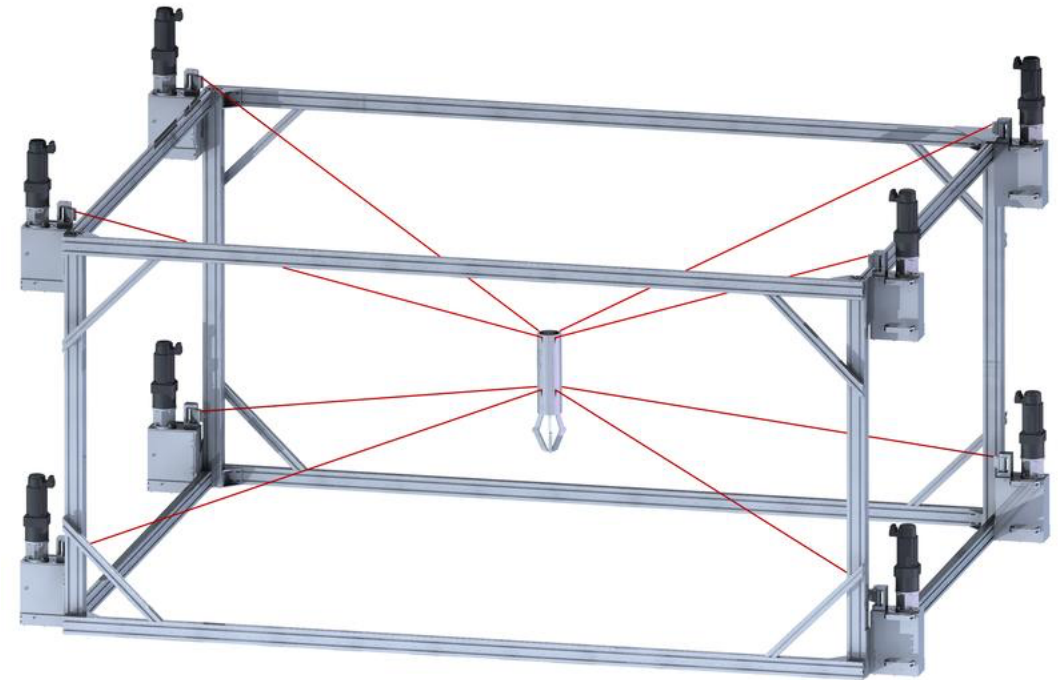


https://doi.org/10.1007/978-3-319-49058-8_60

- Actuation redundancy
 - Serial equivalent?

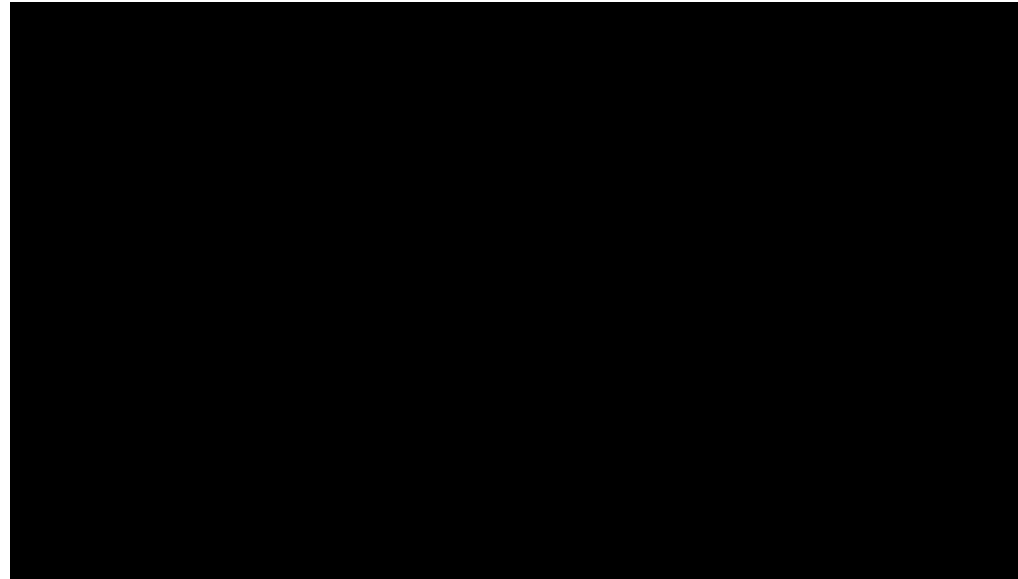


<https://youtu.be/5Erebm0G44I>



IPAnema robot

- Kinematic redundancy
 - Serial equivalent?



https://youtu.be/_vp1ELEtDN4